

Iron-Oxide-Modified Biochar for Removing Arsenic from Groundwater

PROTOCOL-STAGE CASE STUDY

A real scientific topic used to demonstrate review planning, eligibility criteria, database-specific search design, and auditable workflow status.

CURRENT STATUS

Protocol defined

Draft PubMed query recorded

References imported: 0

Evidence conclusions: none

1. Review protocol

RESEARCH QUESTION

How do iron-oxide modification methods, synthesis conditions, water chemistry, and regeneration cycles affect arsenic adsorption by engineered biochar?

Objective

Systematically map and compare experimental evidence on arsenic removal from water using iron-oxide-modified biochar, with emphasis on preparation, water chemistry, performance, characterization, and regeneration.

Scope framework

MATERIAL / SYSTEM

Iron-oxide-modified biochar tested in aqueous arsenic systems

TREATMENT / EXPOSURE

Biochar iron modification method and arsenic-contaminated water exposure

COMPARATOR

Unmodified biochar, alternative iron phases, or differing synthesis conditions

OUTCOME

Arsenic adsorption capacity, removal efficiency, kinetics, selectivity, and regeneration

SCOPE LIMITS

2000-2026 | English | Primary experimental, peer-reviewed studies

2. Eligibility and search design

INCLUDE

- Iron-modified biochar is evaluated
- Arsenic is tested in an aqueous system
- Adsorption or removal performance is reported
- Experimental methods are described

EXCLUDE

- No arsenic outcome
- Biochar is not iron modified
- Review, commentary, or conference abstract only
- No primary experimental data
- Non-aqueous system

Draft PubMed query

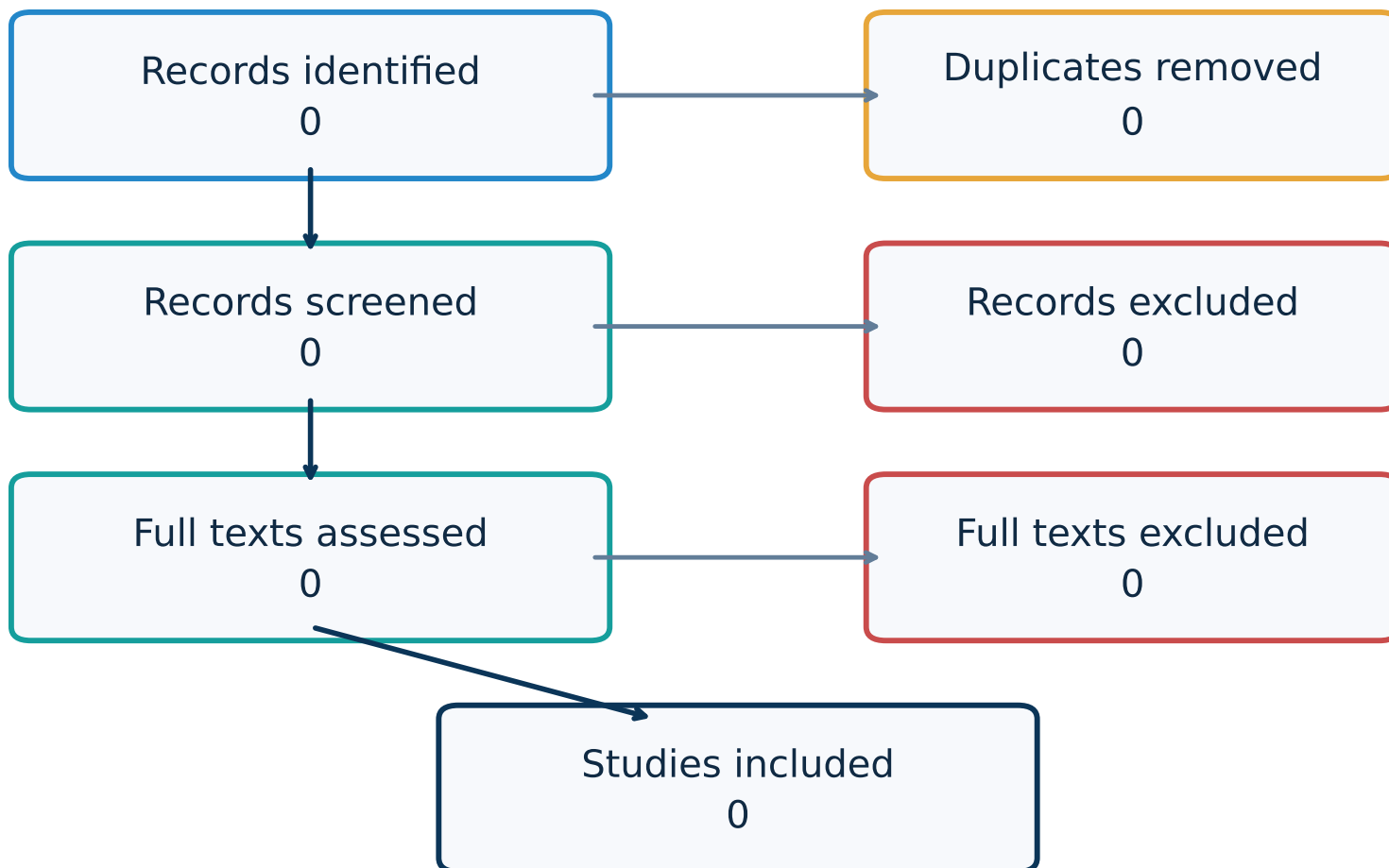
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(biochar OR biocarbon OR "pyrolyzed biomass") AND ("iron oxide" OR magnetite OR hematite OR "zero-valent iron" OR iron-modified) AND (arsenic OR arsenite OR arsenate OR groundwater)
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AUDIT NOTE

This is a protocol-stage draft, not an executed search. Database field tags, search date, and retrieval count must be verified and recorded before screening begins.

3. PRISMA workflow status

Protocol stage - no records imported and no screening decisions recorded



4. Interpretation boundaries and next steps

What this case study demonstrates

- A predefined, chemistry-appropriate research question
- Explicit inclusion and exclusion criteria
- A database-specific Boolean search draft
- A state-derived PRISMA workflow with no invented counts
- A reproducible local project file

What it does not claim

- No database search has yet been executed
- No references have been imported or screened
- No study characteristics or quality judgments have been extracted
- No evidence synthesis or scientific conclusion has been produced

NEXT WORKFLOW STEPS

1. Adapt and execute the query in each selected database.
2. Import legally obtained RIS, BibTeX, CSV, or XLSX exports.
3. Confirm duplicate candidates and screen every active record.
4. Extract reported evidence, assess quality domains, and regenerate the report.